

CS4227_CS5722 - Software Architecture

Component-Based Research Management System

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Assignment 4

Module: Software Architecture

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Summary

Event-Driven Architecture (EDA) is applied to manage research related interactions like discussions, notifications and real-time user engagement in the RMS. This architecture enables flexible and scalable design with asynchronous communication for improved responsiveness and user experience. This assignment describes the main challenges, lessons learned and architectural decisions made during development and future improvements.

Challenges & Lessons Learned

The main challenges during the project were asynchronous event management, real time notifications and time constraints as well as scope changes during continuous reviews. The asynchronous nature of event-driven systems implies natural complexity in terms of event management and state consistency. It proved difficult to demonstrate this complexity because the system did not support simultaneous multi-user interaction. Mock events displaying other user's actions (like tagging or replying) were added to demonstrate the event-driven notifications. These mock events were routed through the event-driven system to model event propagation and notification handling. Also, due to the project's rapid prototyping approach, all features were implemented with TypeScript and static variable databases to reduce development time.

As requirements changed and unexpected technical challenges arose, flexibility became crucial. Time management and team communication were crucial to overcoming doubts about the project scope. Using tools like mitt for event handling made development faster and simpler to create and iterate on. Valuable insights were gained in gathering and understanding user requirements, ensuring the developed features efficiently met user needs.

Architectural Decision Review

The architectural decision review, particularly the scenario-based assessment process, identified several strengths and likely areas for improvement in the Event Driven Architecture implementation. Performance testing showed responsiveness, loading speed and stability which earned it an excellent optimization score. Nevertheless, backend processes including authentication and data storage were missing from comprehensive security assessment. SonarQube discovered risks in security practices including Math.random used for ID generation and recommending randomness handling for security purposes.

Maintainability analysis found numerous bugs mainly associated with code readability, structure and best practices, including SonarQube and ESLint results for variable declarations, operator usage, component management and deprecated elements.

Future Improvements & Self-Assessment

A few areas for improvements are identified. To deliver a strong backend service with persistent storage is crucial for transitioning from prototype to production. Integrated data persistence and security features with databases (such as MongoDB) and user authentication services will solve current challenges.

Another emphasis is on performance improvements such as event processing optimization and frontend rendering latency reduction. Furthermore, accessibility and usability testing - informed design might help improve the user experience further.

Self-assessment: Building modular, responsive frontend applications and asynchronous logic management. Stronger future iterations of the project would be based upon thorough testing, comprehensive error handling and security.